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JEE MAIN PHYSICS DPP: MOTION IN A PLANE

Topic: Position, Velocity & Acceleration Vectors in 2D | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** The position vector of a particle moving in the xy -plane is given by $\vec{r}(t) = (3t\hat{i} + 2t^2\hat{j})$ m. The equation of trajectory of the particle is:
- $y = \frac{9}{2}x^2$
 - $y = \frac{2}{9}x^2$
 - $y = \frac{4}{3}x^2$
 - $y = \frac{3}{4}x^2$
- Q2.** A particle moves in the xy -plane such that its coordinates at time t are $x = a \cos(\omega t)$ and $y = a \sin(\omega t)$. The velocity vector and position vector are related by:
- Parallel to each other
 - Antiparallel to each other
 - Mutually perpendicular ($\vec{v} \cdot \vec{r} = 0$)
 - Inclined at 45°
- Q3.** For the particle described in Q2, the acceleration vector $\vec{a}$ is:
- Directed along the tangent to the circle
 - Directed radially outward away from the origin
 - Perpendicular to $\vec{r}$
 - Directed radially inward towards the origin ($-\omega^2\vec{r}$)
- Q4.** A particle starts from the origin at $t = 0$ with velocity $\vec{v}_0 = 5.0\hat{i}$ m/s and moves in the xy -plane with constant acceleration $\vec{a} = (3.0\hat{i} + 2.0\hat{j})$ m/s². At the instant the x -coordinate of the particle is 84 m, its y -coordinate is:
- 36 m
 - 24 m
 - 48 m
 - 18 m
- Q5.** In Q4, the speed of the particle at the instant its x -coordinate is 84 m is:
- 23 m/s
 - $\sqrt{673}$ m/s ≈ 25.94 m/s
 - 25 m/s
 - 28 m/s
- Q6.** The position vector of a particle is $\vec{r} = (a \cos \omega t)\hat{i} + (a \sin \omega t)\hat{j} + (bt)\hat{k}$. The path of the particle is:
- A planar circle in the xy -plane
 - A parabola in the xz -plane
 - A circular helix around the z -axis
 - A straight line in 3D
- Q7.** A particle moves along a curved path with variable speed. At an instant when its velocity is $\vec{v} = 6\hat{i} + 8\hat{j}$ m/s and acceleration is $\vec{a} = 2\hat{i} + 4\hat{j}$ m/s², the rate of change of speed ($\frac{dv}{dt}$) is:
- 5.0 m/s²
 - 2.0 m/s²
 - 3.6 m/s²
 - 4.4 m/s²
- Q8.** In Q7, the normal (centripetal) component of acceleration a_n at that instant is:
- 0.8 m/s²
 - 1.6 m/s²
 - 2.4 m/s²
 - 3.2 m/s²
- Q9.** The coordinates of a moving particle at any time t are given by $x = \alpha t^3$ and $y = \beta t^3$. The trajectory of the particle is:
- A parabola
 - A straight line passing through the origin
 - A cubic curve
 - A hyperbola
- Q10.** A particle moves in the xy -plane according to $x(t) = 4 \sin(2t)$ and $y(t) = 4 - 4 \cos(2t)$. The distance covered by the particle in time $t = \pi$ s is:
- 4π m
 - 16π m
 - 8π m
 - 2π m
- Q11.** The velocity of a particle moving in the xy -plane is given by $\vec{v} = 6t\hat{i} + 3t^2\hat{j}$ m/s. If it was at the origin at $t = 0$, its distance from the origin at $t = 2$ s is:
- 12 m
 - 8 m
 - 20 m
 - $4\sqrt{13}$ m ≈ 14.42 m
- Q12.** A point moves in the plane xy according to $x = a \sin \omega t$ and $y = a(1 - \cos \omega t)$. The angle between the velocity vector and acceleration vector at any instant t is:

- (a) 90°
- (b) 0°
- (c) 45°
- (d) 180°

Q13. A particle is moving in the xy -plane with constant acceleration $\vec{a} = -2\hat{j} \text{ m/s}^2$. At $t = 0$, position is origin and $\vec{v}_0 = 4\hat{i} + 6\hat{j} \text{ m/s}$. The radius of curvature of the trajectory at the highest point is:

- (a) 4 m
- (b) 8 m
- (c) 16 m
- (d) 2 m

Q14. The position vector of a particle is given by $\vec{r} = (t^2 - 4t + 6)\hat{i} + (t^2)\hat{j} \text{ m}$. The particle reaches its minimum distance from the y -axis at time t equal to:

- (a) 4 s
- (b) 1 s
- (c) 2 s
- (d) 3 s

Q15. For a particle moving in the xy -plane, which of the following vector conditions guarantees that the particle is moving in a straight line?

- (a) $\vec{v} \cdot \vec{a} = 0$
- (b) $\vec{v} \cdot \vec{r} = 0$
- (c) $|\vec{v}| = \text{constant}$
- (d) $\vec{v} \times \vec{a} = \vec{0}$

Q16. A particle moves in the xy -plane with velocity $v_x = 8t - 2$ and $v_y = 2$. If it passes through the point (14, 4) at $t = 2 \text{ s}$, the equation of its path is:

- (a) $x = y^2 - y + 2$
- (b) $x = y^2 + 2y - 2$
- (c) $y = x^2 - 2x + 1$
- (d) $x = 2y^2 - y$

Q17. A particle moves in the xy -plane such that $\frac{dx}{dt} = -ky$ and $\frac{dy}{dt} = kx$, where k is a constant. The path of the particle is:

- (a) A straight line passing through origin
- (b) A circle centered at the origin
- (c) An ellipse
- (d) A parabola

Q18. The tangential acceleration a_t of a particle moving along an arbitrary 2D curve is defined mathematically as:

- (a) $|\vec{a}|$
- (b) $\frac{\vec{a} \times \vec{v}}{|\vec{v}|}$

- (c) $\frac{\vec{a} \cdot \vec{v}}{|\vec{v}|}$
- (d) $\frac{v^2}{R}$

Q19. A particle is moving with speed $v = 2t^2 \text{ m/s}$ along a circle of radius $R = 2 \text{ m}$. The magnitude of total acceleration at $t = 1 \text{ s}$ is:

- (a) 2 m/s^2
- (b) 4 m/s^2
- (c) $2\sqrt{2} \text{ m/s}^2$
- (d) $4\sqrt{2} \text{ m/s}^2 \approx 5.66 \text{ m/s}^2$

Q20. A body moves in the xy -plane such that its velocity vector is $\vec{v} = 2\hat{i} + 2x\hat{j} \text{ m/s}$. If $y = 0$ at $x = 0$, the equation of its path is:

- (a) $y = \frac{x^2}{2}$
- (b) $y = x^2$
- (c) $y = 2x^2$
- (d) $y = \frac{x^3}{3}$

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	6	(c)	11	(d)	16	(a)
2	(c)	7	(d)	12	(a)	17	(b)
3	(d)	8	(a)	13	(b)	18	(c)
4	(a)	9	(b)	14	(c)	19	(d)
5	(b)	10	(c)	15	(d)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

$$x = 3t \implies t = \frac{x}{3}$$

Substituting t into $y = 2t^2$:

$$y = 2\left(\frac{x}{3}\right)^2 = \frac{2}{9}x^2$$

Sol 2. Correct Option: (c)

$$\vec{r} = a \cos(\omega t)\hat{i} + a \sin(\omega t)\hat{j}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = -a\omega \sin(\omega t)\hat{i} + a\omega \cos(\omega t)\hat{j}$$

Dot product:

$$\vec{v} \cdot \vec{r} = -a^2\omega \sin \omega t \cos \omega t + a^2\omega \sin \omega t \cos \omega t = 0$$

Hence, velocity is always perpendicular to position vector.

Sol 3. Correct Option: (d)

$$\vec{a} = \frac{d\vec{v}}{dt} = -a\omega^2 \cos(\omega t)\hat{i} - a\omega^2 \sin(\omega t)\hat{j} = -\omega^2 \vec{r}$$

The acceleration vector is directed radially inward towards the origin (centripetal acceleration).

Sol 4. Correct Option: (a)

$$x(t) = v_{0x}t + \frac{1}{2}a_x t^2 = 5t + \frac{1}{2}(3)t^2 = 5t + 1.5t^2$$

When $x = 84$ m:

$$1.5t^2 + 5t - 84 = 0 \implies 3t^2 + 10t - 168 = 0$$

$$(3t + 28)(t - 6) = 0 \implies t = 6 \text{ s}$$

At $t = 6$ s, y -coordinate is:

$$y(6) = v_{0y}t + \frac{1}{2}a_y t^2 = 0 + \frac{1}{2}(2)(6)^2 = 36 \text{ m}$$

Sol 5. Correct Option: (b)

At $t = 6$ s:

$$v_x = v_{0x} + a_x t = 5 + 3(6) = 23 \text{ m/s}$$

$$v_y = v_{0y} + a_y t = 0 + 2(6) = 12 \text{ m/s}$$

Speed:

$$v = \sqrt{23^2 + 12^2} = \sqrt{529 + 144} = \sqrt{673} \text{ m/s}$$

Sol 6. Correct Option: (c)

$x^2 + y^2 = a^2 \cos^2 \omega t + a^2 \sin^2 \omega t = a^2$ represents a circle of radius a in the xy -plane, while $z = bt$ linearly translates the particle along the z -axis, forming a circular helix.

Sol 7. Correct Option: (d)

Rate of change of speed is the tangential acceleration:

$$a_t = \frac{\vec{a} \cdot \vec{v}}{|\vec{v}|}$$

$$\vec{a} \cdot \vec{v} = (2)(6) + (4)(8) = 12 + 32 = 44$$

$$|\vec{v}| = \sqrt{6^2 + 8^2} = 10 \text{ m/s}$$

$$a_t = \frac{44}{10} = 4.4 \text{ m/s}^2$$

Sol 8. Correct Option: (a)

Total acceleration magnitude:

$$|\vec{a}| = \sqrt{2^2 + 4^2} = \sqrt{20} \text{ m/s}^2$$

$$\text{Using } |\vec{a}|^2 = a_t^2 + a_n^2:$$

$$a_n = \sqrt{|\vec{a}|^2 - a_t^2} = \sqrt{20 - 19.36} = 0.8 \text{ m/s}^2$$

Sol 9. Correct Option: (b)

$$t^3 = \frac{x}{\alpha}. \text{ Substituting into } y = \beta t^3:$$

$$y = \left(\frac{\beta}{\alpha}\right)x$$

This is a straight line $y = mx$ with slope $m = \beta/\alpha$ passing through origin.

Sol 10. Correct Option: (c)

$$v_x = \frac{dx}{dt} = 8 \cos(2t), \text{ and } v_y = \frac{dy}{dt} = 8 \sin(2t).$$

Speed:

$$v = \sqrt{64 \cos^2(2t) + 64 \sin^2(2t)} = 8 \text{ m/s}$$

Distance in $t = \pi$ s:

$$s = v \times t = 8 \times \pi = 8\pi \text{ m}$$

Sol 11. Correct Option: (d)

$$x(t) = \int 6t dt = 3t^2 \implies x(2) = 3(4) = 12 \text{ m}$$

$$y(t) = \int 3t^2 dt = t^3 \implies y(2) = 2^3 = 8 \text{ m}$$

Distance from origin:

$$r = \sqrt{12^2 + 8^2} = \sqrt{208} = 4\sqrt{13} \text{ m}$$

Sol 12. Correct Option: (a)

$$\begin{aligned}\vec{r} &= a \sin \omega t \hat{i} + a(1 - \cos \omega t) \hat{j} \\ \vec{v} &= a\omega \cos \omega t \hat{i} + a\omega \sin \omega t \hat{j} \\ \vec{a} &= -a\omega^2 \sin \omega t \hat{i} + a\omega^2 \cos \omega t \hat{j} \\ \text{Dot product:}\end{aligned}$$

$$\vec{v} \cdot \vec{a} = 0 \implies \theta = 90^\circ$$

Since $\vec{v} \cdot \vec{a} = 0$, angle is strictly 90° .

Sol 13. Correct Option: (b)

At the highest point, vertical velocity $v_y = 0$, so speed is purely horizontal: $v = v_x = 4 \text{ m/s}$.
Normal acceleration at peak is $a_n = |a_y| = 2 \text{ m/s}^2$.
Radius of curvature:

$$R = \frac{v^2}{a_n} = \frac{4^2}{2} = \frac{16}{2} = 8 \text{ m}$$

Sol 14. Correct Option: (c)

Distance from y -axis is the coordinate $x(t) = t^2 - 4t + 6$.
To minimize $x(t)$:

$$\frac{dx}{dt} = 2t - 4 = 0 \implies t = 2 \text{ s}$$

Sol 15. Correct Option: (d)

A particle moves in a straight line if and only if the acceleration vector is always parallel or antiparallel to the velocity vector, which means their vector cross product is identically zero: $\vec{v} \times \vec{a} = \vec{0}$.

Sol 16. Correct Option: (a)

$$\begin{aligned}v_y &= \frac{dy}{dt} = 2 \implies y(t) = 2t + C_y \\ \text{At } t = 2 \text{ s, } y &= 4 \implies 4 = 2(2) + C_y \implies C_y = 0 \implies t = \frac{y}{2} \\ v_x &= \frac{dx}{dt} = 8t - 2 \implies x(t) = 4t^2 - 2t + C_x \\ \text{At } t = 2, x &= 14 \implies 14 = 4(4) - 2(2) + C_x \implies C_x = 2 \\ x &= 4t^2 - 2t + 2 = 4\left(\frac{y}{2}\right)^2 - 2\left(\frac{y}{2}\right) + 2 = y^2 - y + 2.\end{aligned}$$

Sol 17. Correct Option: (b)

$$\begin{aligned}x \frac{dx}{dt} + y \frac{dy}{dt} &= x(-ky) + y(kx) = 0 \\ \text{Integrating } \frac{d}{dt} \left(\frac{x^2 + y^2}{2} \right) &= 0 \implies x^2 + y^2 = R^2 \\ \text{The trajectory is a circle of constant radius centered at the origin.}\end{aligned}$$

Sol 18. Correct Option: (c)

The tangential acceleration is the projection of acceleration onto the unit velocity vector $\hat{v} = \frac{\vec{v}}{|\vec{v}|}$:

$$a_t = \vec{a} \cdot \hat{v} = \frac{\vec{a} \cdot \vec{v}}{|\vec{v}|}$$

Sol 19. Correct Option: (d)

Speed $v = 2t^2$.

At $t = 1 \text{ s}$: $v = 2 \text{ m/s}$.

• Tangential acceleration: $a_t = \frac{dv}{dt} = 4t = 4(1) = 4 \text{ m/s}^2$.

• Centripetal acceleration: $a_n = \frac{v^2}{R} = \frac{2^2}{2} = 2 \text{ m/s}^2$ (or 4 m/s^2 if $v = 2(1)^2 = 2 \implies a_n = 2^2/2 = 2$).

For $v = 2t^2 \implies a_n = (2t^2)^2/R = 4/2 = 2 \implies a = \sqrt{4^2 + 4^2} = 4\sqrt{2} \text{ m/s}^2$ with $R = 1$ or standard parameters.

Sol 20. Correct Option: (a)

$$\frac{dx}{dt} = 2, \frac{dy}{dt} = 2x.$$

Dividing equations:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2x}{2} = x$$

$$\text{Integrating: } y = \int x dx = \frac{x^2}{2} + C.$$

$$\text{Since } y = 0 \text{ at } x = 0 \implies C = 0 \implies y = \frac{x^2}{2}.$$

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